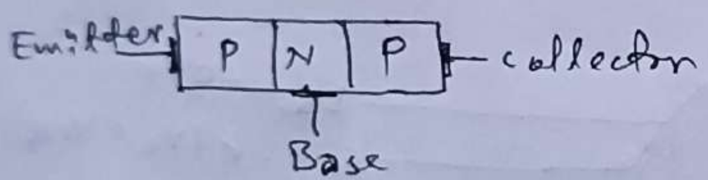
Bipolar junction Transistor [BJT] :-

Transistor is a three terminal electronic device formed by the combination of two P-N junctions in a specific manner. The three terminals are known as collector, base and emitter. A general P-N junction diode has low resistance on forward biased case and very high value of resistance on reverse bias. So, a device with two P-N junctions in which a low signal is applied to of forward junction and a large power gain is obtained across a next reverse bias junction is a transistor. A transistor is named after transformation of resistor from low value to high value resistance.

A transistor consists of two junctions at different biasing mode and both holes and electrons are involved to conduct current, it is also named as bipolar junction transistor. Hence, a transistor is mainly use to amplify or multiply input current, voltage or power in circuit.
There are two types of bipolar junction transistor.

1. P-N-P transistor:-

A PNP transistor consists of two P-type semiconductor sand witched together with thin layer of N-type between them. A hole is major charge carrier in PNP transistor. An arrow head is drawn on the emitter to distinguish it from collector, the direction is pointing towards direction of flow of positive charge i.e. holes from P to N type.



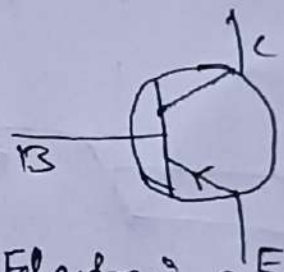
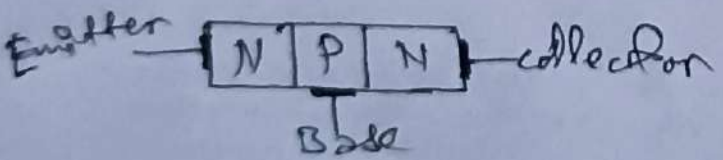


fig:-Electronic symbol of PNP transis

2. N-P-N transistor:-

A NPN transistor consists of two N-type semiconductor sand witched together with thin layer of P-type between them. An electron is major charge carrier in NPN transistor. A arrow head is drawn on the emitter to distinguish it from collector, the direction is pointing towards direction of flow of positive charge (direction of conventional current) i.e. holes from P to N type. It is use widely as electron is major charge carriers in it and electron has larger mobility compare to holes.



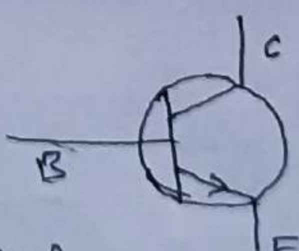


fig: Electronic symbol of NPN transistor

Terminals of transistors:-

A transistors is three terminal electronic device which are as follow:-

1. Emitter:-

It is on left hand section of transistor which is thick layers and is highly doped region. Emitter acts as source of majority charge carriers (i.e. holes in PNP transistor and electrons in case of NPN transistor) in transistors.

2. Base:-

The middle section of transistor is called base. This is very lightly doped and very thin of about 10^{-6} m as compared to emitter and collector. The base region in transistor acts as a pass way for injected charge carriers from emitter to collector.

3. Collector:-

It lies at right hand side of transistor and is most thicker region, it is moderately doped region. The main function of collector is to collect majority charge carriers through base.

Note: The resistance between Base to emitter terminal is less than that of resistance between collector to base.

When Semiconductor layers are put together, majority carriers will flow across junctions, creating contact of potential differences between emitter and base and between collector and base. In the absence of external bias potential, the potential of the N side of a P-N junction is higher than that of P-side by contact potential V_C. The variation in potential in three sections of NPN transistor are shown in figure.

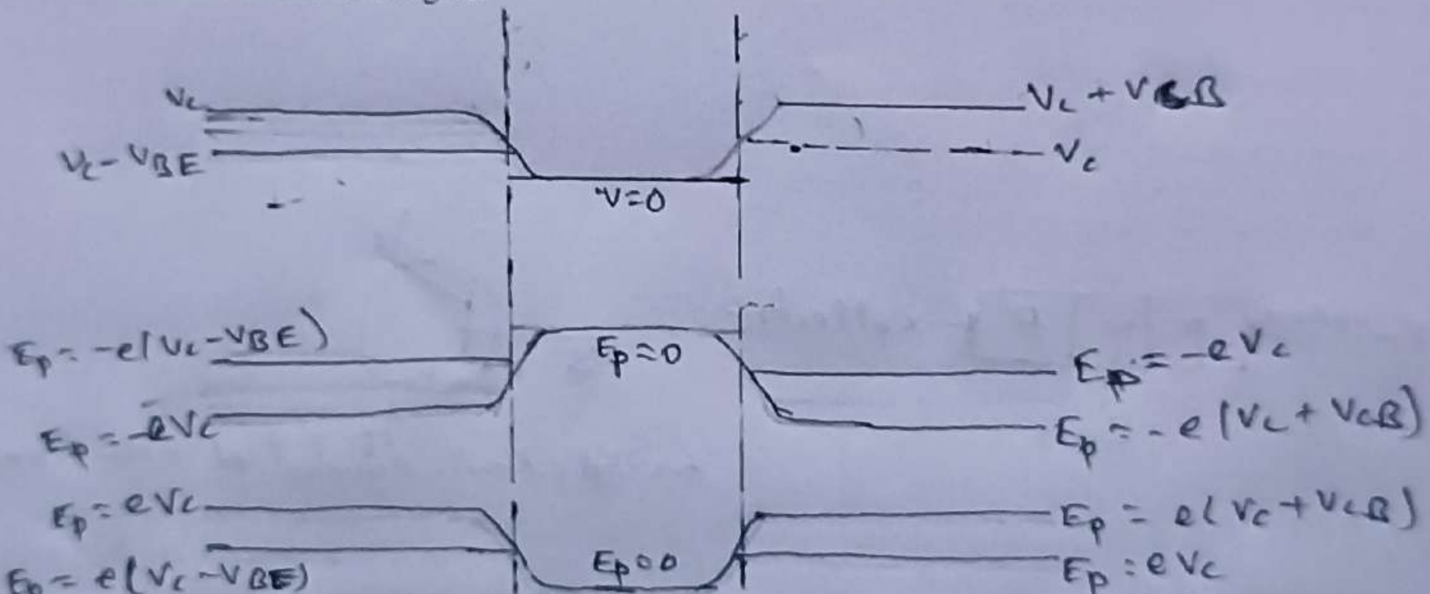
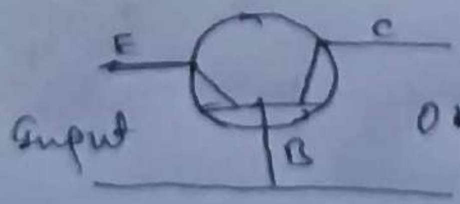


fig:- potential curve & band structure in transistor.

Different Configuration or mode of connection of transistor:-

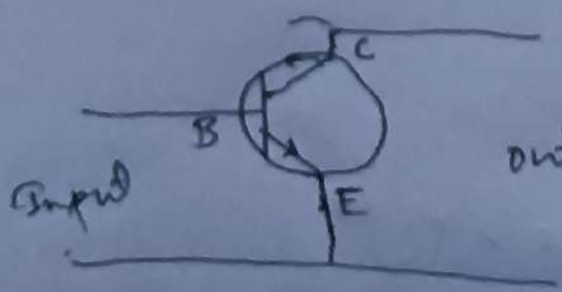
1. Common base Configuration:-

In this configuration base is made common terminal to emitter and collector. The input is applied between emitter and base terminal where output is taken from collector to base terminals.



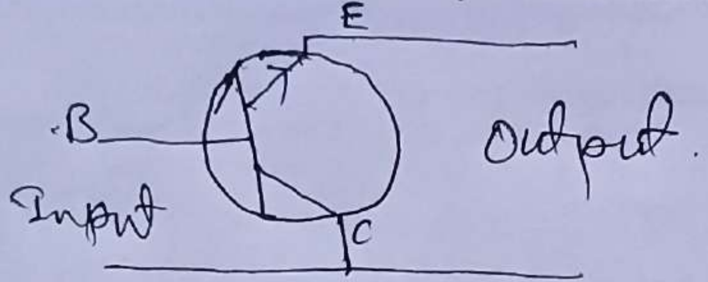
2. Common emitter Configuration:-

In this configuration emitter is made common terminal to base and collector. The input is applied between emitter and base where output is taken from emitter and collector.



3. Common collector Configuration:-

In this configuration collector is made common terminal to base and emitter. The input is applied between base and collector where output is taken from emitter and collector.



Characteristics of a transistor in common Base Configuration:-

In Common base configuration of a transistor, base is made common to both emitter and collector terminals. Here, emitter base junction is forward bias with a power source V_{EE} and collector base junction is reverse biased by a power source V_{CC}.

The supply power V_{EE} increase the potential of P-type base relative to N-type emitter by an amount V_{BE}. So, the potential difference between emitter and base is reduced to V_C- V_{BE} . Similarly, V_{CC} lower down potential of base relative to the collector by an amount V_{EB} . So, the potential difference between emitter and collector is increased to $V_C + V_{CB}$.

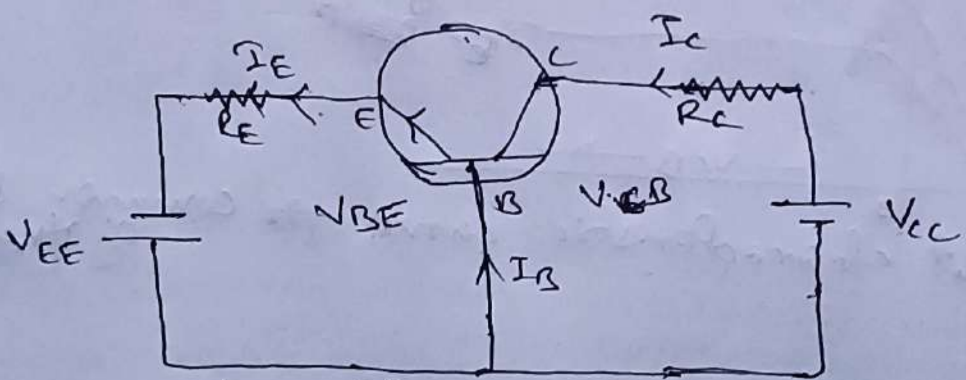


fig:- NPN transistor in common Base mode

The electrical behavior of a transistor in different mode is given by various current and voltage graphs. The important characteristics of CB configuration is given as

a. Input characteristics:-

The input characteristics of common base configuration of transistor is obtained by plotting input emitter current (I_E) versus input base emitter potential (V_{BE}) at constant output collector base voltage (V_{CB}).

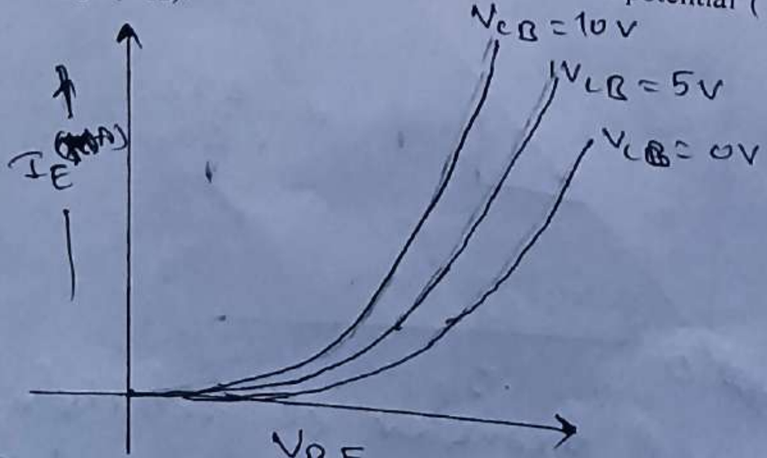


fig:- Input characteristic curve for common Base mode of transistor

1. The emitter current I_E increases rapidly on small increases in emitter base voltage (V_{BE}). This signify small input resistance.
2. The emitter current is almost independent of collector base voltage (V_{CB}).
3. The emitter current is small and finite even when emitter base voltage is about zero for given value of V_{CB}.
4. The input resistance is given by, $r_i = \frac{(\Delta V_{BE})}{(\Delta I_E)}$ at V_{CB} constant. As resistance changes from point to point, it is also called dynamic resistance and is of order of few ohms.

b. Output characteristics:-

The output characteristics is obtained by plotting collector current (I_C) versus collector base voltage (V_{CB}) at constant emitter current (I_E).

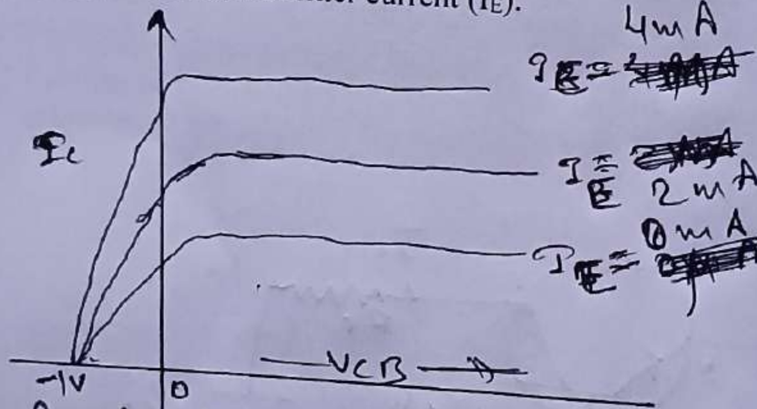


fig:- output characteristic curve for common Base mode of transistor.

1. The collector current varies largely with V_{CB} only at low voltage less than 1 volt.
2. The collector current is almost constant with V_{CB} at more than 1 Volt.
3. The large change in V_{CB} result to small change in I_C, signify high output resistance i.e.

$$r_0 = \frac{(\Delta V_{CB})}{(\Delta I_C)} \text{ at } I_E = \text{constant}$$

It is constant almost and of order of several kilo-ohms.

c. Current transfer characteristics:-

A plot between output collector current (I_C) versus input emitter current (I_E) gives current transfer characteristics for CB-mode transfer. The relation is linear in nature. The slope of line gives current transfer factor in CB-mode as

$$\alpha = \frac{(\Delta I_C)}{(\Delta I_E)}$$

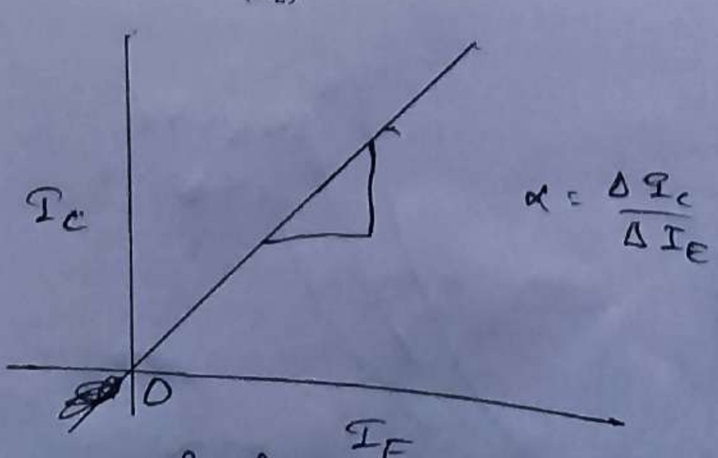


fig:- current transfer characteristic curve in Common Base mode transistor.

Characteristics of a transistor in common Emitter Configuration:-

In Common emitter configuration of a transistor, emitter is made common to both base and collector terminals. Here, emitter base junction is forward bias with a power source V_{BB} and collector base junction is reverse biased by a power source V_{CC}.

The supply power V_{BB} increase the potential of P-type base relative to N-type emitter by an amount V_{BE}. So, the potential difference between emitter and base is reduced to $V_C - V_{BE}$. Similarly, V_{CC} lower down potential of base relative to the collector by an amount V_{CB}. So, the potential difference between emitter and collector is increased to $V_C + V_{CB}$.

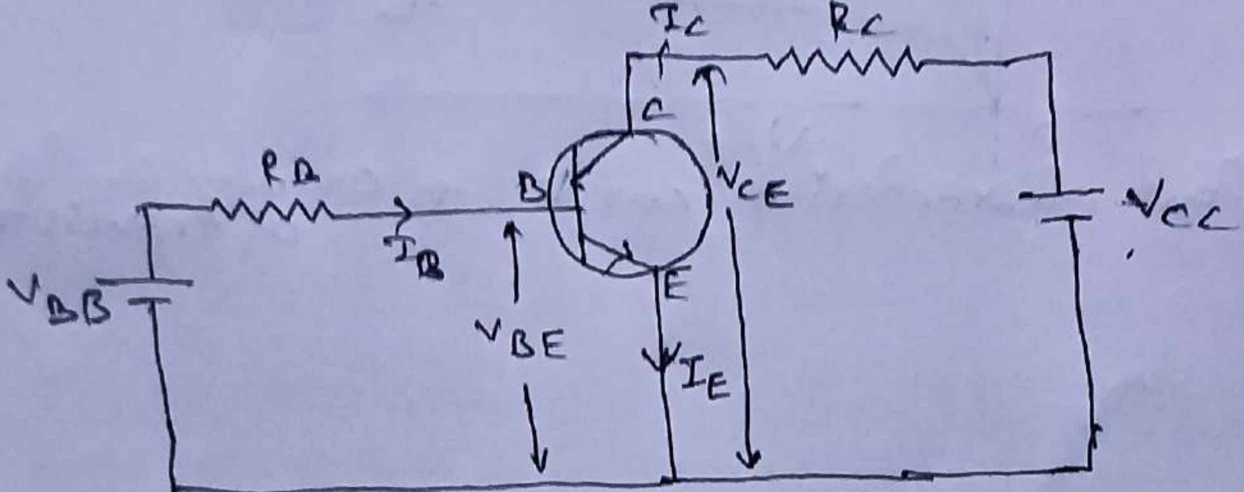


fig: NPN transistor in common Emitter mode.

The electrical behavior of a transistor in different mode is given by various current and voltage graphs. The important characteristics of CE configuration is given as

a. Input characteristics:-

The input characteristics of common emitter configuration of transistor is obtained by plotting input base current (I_B) versus input base emitter potential (V_{BE}) at constant output collector emitter voltage (V_{CE}).

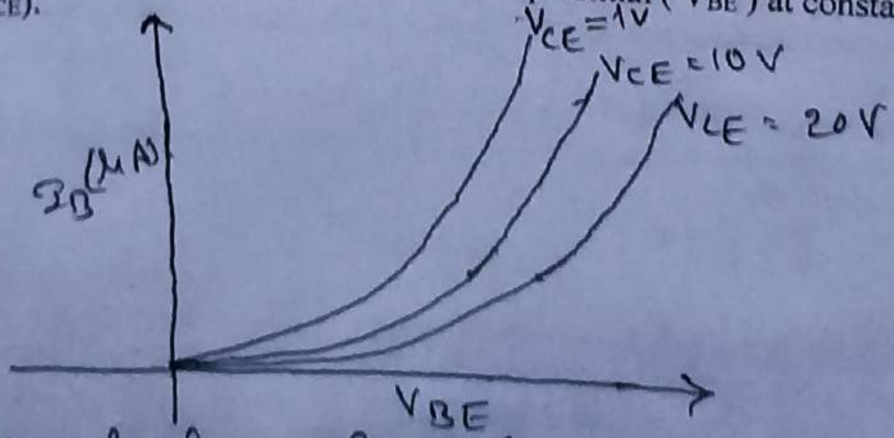


fig:- Input characteristics curve for common emitter mode of transistor.

1. The base current (I_B) is zero below V_{BE} equal to 0.3V for 'Ge' and 0.7 V for 'Si', it starts to increase only after V_{BE} equal to 0.3 V or 0.7 V.
2. Base current increases with increase in collector emitter voltage (V_{CE}).
3. The input resistance is given by, $r_i = \frac{(\Delta V_{BE})}{(\Delta I_B)}$ at V_{CE} constant. As resistance changes from point to point, it is also called dynamic resistance and is of small order.

b. Output characteristics:-

The output characteristics is obtained by plotting collector current (I_C) versus collector emitter voltage (V_{CE}) at constant base current (I_B).

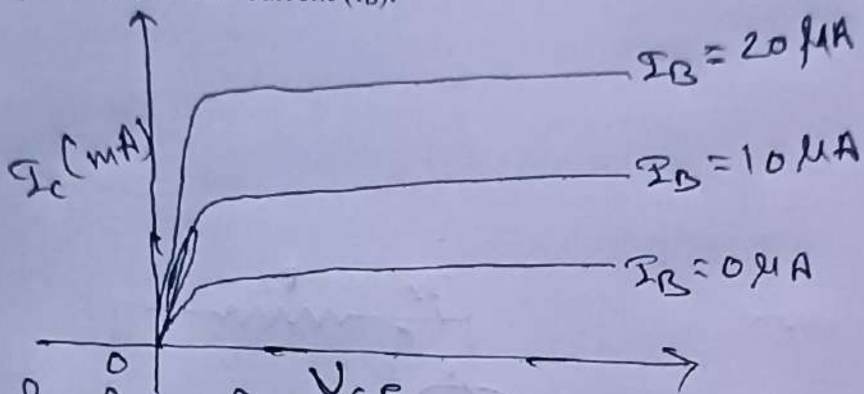


fig:- output characteristic curve for common emitter mode of transistor

1. The collector current increases rapidly for V_{CE} less than 1 volt.
2. The collector current (I_c) is almost constant for V_{CE} more than 1 Volt. This is known as saturation current.
3. There is absent of collector current for below $I_B = 0\ \mu A$ called cutoff region.
4. The large change in V_{CE} result to small change in I_C, signify high output resistance i.e.

$$r_0 = \frac{(\Delta V_{CE})}{(\Delta I_C)} \text{ at } I_B = \text{constant}$$

It is order of several kilo-ohms.

c. Current transfer characteristics:-

A plot between output collector current (I_C) versus input base current (I_B) gives current transfer characteristics for CE-mode transistor. The relation is linear in nature. The slope of line gives current transfer factor in CB-mode as

$$\beta = \frac{(\Delta I_C)}{(\Delta I_B)}$$

Relation between 'α' and 'β' in transistor:-

The relation between current transfer factor in common base mode (**α**) and in common emitter mode (**β**) is

The currents in transistor is related as

$$I_E = I_B + I_C$$

Diving each side by I_C,

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + 1$$

$$\text{or, } \frac{1}{\alpha} = \frac{1}{\beta} + 1 \text{.......................} \ 1$$

$$\text{or, } \frac{1}{\alpha} = \frac{1+\beta}{\beta}$$

$$\text{or, } \alpha = \frac{\beta}{1+\beta} \text{ .... } 2$$

Again, from eq. 1

$$\frac{1}{\beta} = \frac{1}{\alpha} - 1$$

$$\text{or, } \frac{1}{\beta} = \frac{1-\alpha}{\alpha}$$

or, $\beta = \frac{\alpha}{1-\alpha}$ 3

Eqn 1, 2 and 3 are required relations between 'α' and 'β'.

Transistor as a voltage amplifier:-

The most important application of transistor is it is use as an amplifier.

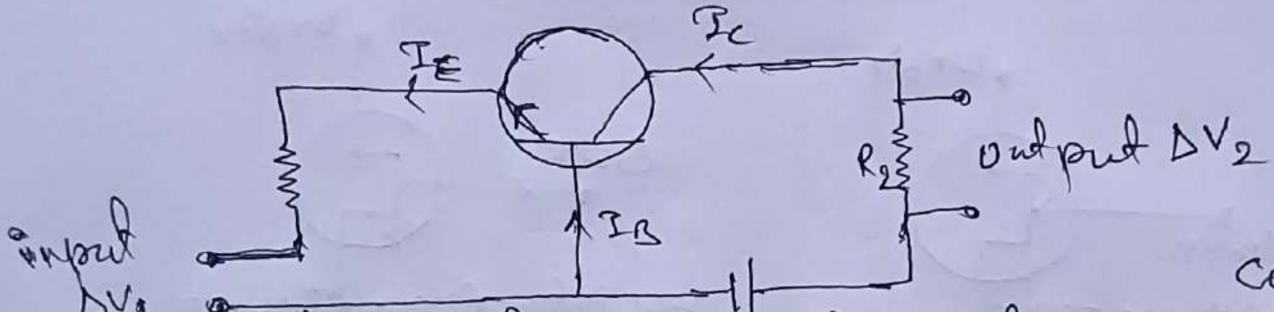


fig:- amplier mode of transistor in common base configuration

For, ΔV_1 be input voltage applied in CB-mode configuration of transistor and ΔV_2 be its corresponding output voltage. 'R_1' and 'R_2' be input and output resistances respectively. For V_1 and V_2 be biasing potential for B-E junction in forward direction and B-C junction in reverse biased respectively then,

Output voltage $\Delta V_2 = R_2 \,.\, \Delta I_C$ 1

from current transfer characteristic;

for common base, current gain factor is $\alpha = \frac{(\Delta I_C)}{(\Delta I_E)}$

$$\text{or,} (\Delta I_C) = \alpha(\Delta I_E)$$

$$\text{or, } (\Delta I_C) = \alpha \frac{\Delta V_1}{(R_1 + R)} \text{ .... } 2$$

where 'R' is resistance between emitter – base junction.

Then, eqn 1, becomes with eqn 2; $\Delta V_2 = \alpha \frac{R_2}{(R_1+R)} \Delta V_1$

Here, as 'R_2' is output resistance which has very large value of order several kilo-ohms in compare to input resistance R_1

So, $R_2 >> (R_1 + R)$ then this implies, $\Delta V_2 >> \Delta V_1$ i.e. output voltage is larger than input voltage. Hence, voltage is amplified by a transistor and is use widely as voltage amplifier electronic device.

Field effect transistor (FET):-

It is a three terminal semiconductor device in which current conduction is due to one type of charge carriers i.e. either by holes or electrons and current through it is controlled by the effect of applied electric field. The FET has high input impedance and low noise level. FET are of two types;

a. Junction Field Effect Transistor (JFET)
b. Metal Oxide Semiconductor Field Effect Transistor (MOSFET)

a. Junction Field Effect Transistor (JFET):

A JFET consists of a p-type and n-type silicon bar containing two p-n junctions at the sides. The bar forms the conducting channel for the charge carriers. When the bar JFET is of n-type, it is called n-channel JFET and when it is of p-type it is called p-channel JFET as shown in fig. The two p-n junctions that make up the diodes are connected internally and the common terminal is called a gate. Others two terminal of JFET are source and drain taken out from the bar. Thus, a JFET has essentially three terminals Viz. gate(G), Source (S) and drain (D).

The electronic symbol of JFET is shown in fig below:

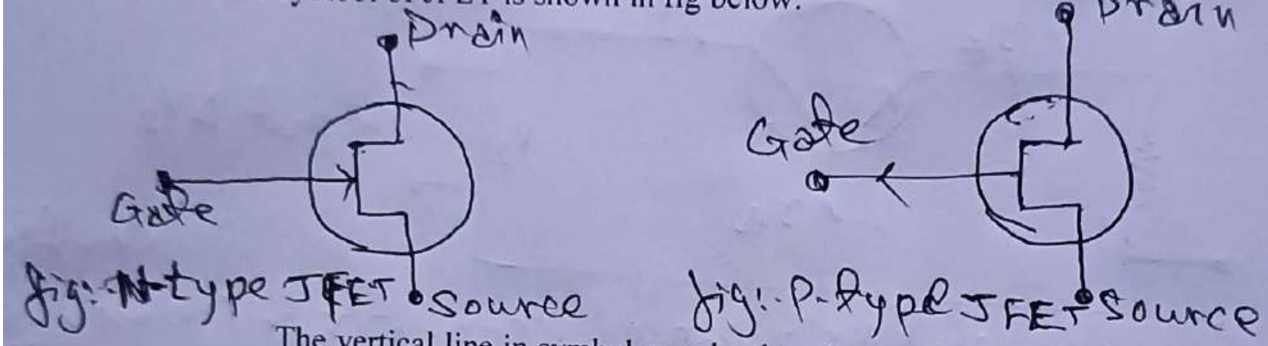


fig: N-type JFET   fig: P-type JFET

The vertical line in symbol may be thought as channel in which source and drain are connected to this line. For the channel of JFET is n-type or p-type, the arrow on the gate is given as shown in fig. In n-channel it is directing inwards while in p-channel it is facing outward. Drain and source are symmetrical showing that they are interchangeable.

Working:

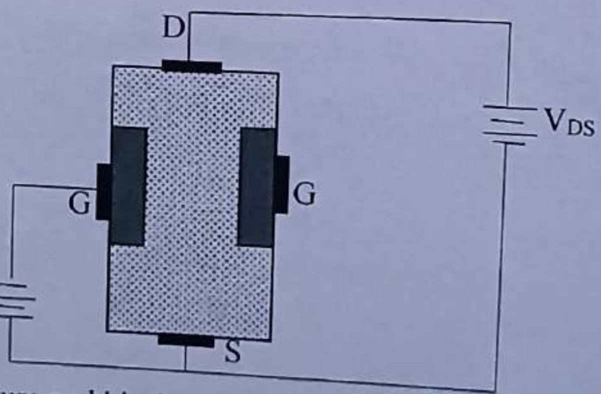


Fig: Structure and biasing of N-channel type JEFET.

i. When voltage V_{DS} is applied b/w drain and source terminals and voltage on the gate is zero as shown in fig. then the two p-n junctions at the sides of the bar establish depletion layers. The electron will flow from source to drain through a channel between the depletion layers. The size of these layers determined the width of the channel and hence the current conduction through the bar.

ii. When a reverse voltage V_{GS} is applied the gate and source in fig, the width of the depletion layers is increased. This reduces the width of the conducting channels, thereby increasing the resistance of n-type bar. Thus, the current from source to drain is decreased. Also, if the reverse voltage on the gate is decreased the width of the depletion layers also decreases. This increased the width of the conducting channel and hence source to drain current.

Thus, this conclude that the current from source to drain can be controlled by the application of potential (i.e. electric field) on the gate. Due to this, the device is called field effect transistor. Hence, a JFET is always used in reverse potential of Gate to source voltage. The p-channel JFET works in same manner except that channel current carriers will be the holes instead of electrons and the polarities of V_{GS} and V_{DS} are reversed.

Output characteristics of JFET:

The output characteristics of the JFET is a curve drawn between drain current and drain source voltage of a JFET at constant gate- source voltage. The drain source voltage above which drain current is maximum and no more increase in it for particular value of V_{GS} and gets saturated, this voltage is called pinch off voltage. On increasing the V_{GS} the channel size reduced and eventually, the drain current reduced to zero, then this particular value of reverse V_{GS} potential is known as cutoff voltage.

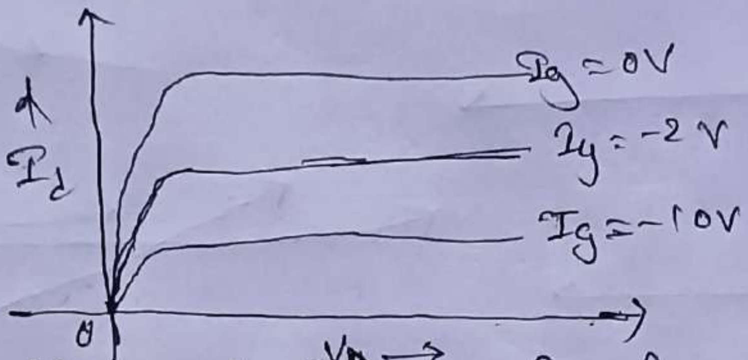


fig:- out put characteristics of JFET.

Advantages of JFET:

A JFET is a voltage controlled, constant current device in which variations in input voltage control the output current. Some of advantages of JFET are;

i. It has a very high input impedance (100MΩ). This permits high degree of isolation between input and output circuits.
ii. The operation of a JFET depends upon the bulk material current carriers tat do not cross junction.
iii. A JFET has a negative temperature coefficient of resistance. This avoids the risk of terminal runway.
iv. A JFET has a very high-power gain. This eliminates the necessity of using driver stage.
v. A JFET has a smaller size, longer life and high efficiency.

b. Metal Oxide Semiconductor Field Effect Transistor (MOSFET) :

MOSFET is a FET that consists of metal oxide semiconductor region for field effect on current conduction. It's one of terminal is of metal conducting type with an insulator (SiO_2) between others two layers. It is of two types Depletion MOSFET (DEMOSFET) and Enhanced MOSFET (EMOSFET). Both are further available with P-channel and N-channel MOSFET. As a JEFET can only work in case of reverse Gate to source voltage, a depletion MOSFET works in both forward and reverse potential of gate to source voltage and enhanced MOSFET works in only in forward V_{GS} voltage. The electronic symbol of MOSFET is as shown in fig;

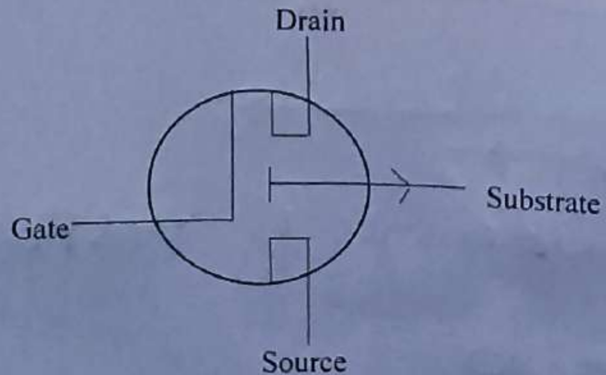


Fig: P-channel MOSFET

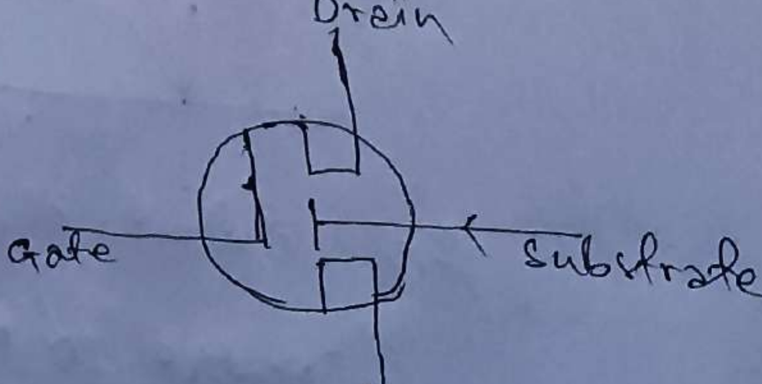


Fig:. N-channel MOSFET

A P-channel MOSFET consists of two heavily doped p-type region in a N-type substrate where as N-channel MOSFET consists of two heavily doped n-type region in a P-type substrate. These heavily doped region acts as drain and source. A metal layer over an insulating layer silicon dioxide above substrate acts as a gate as shown in figure.

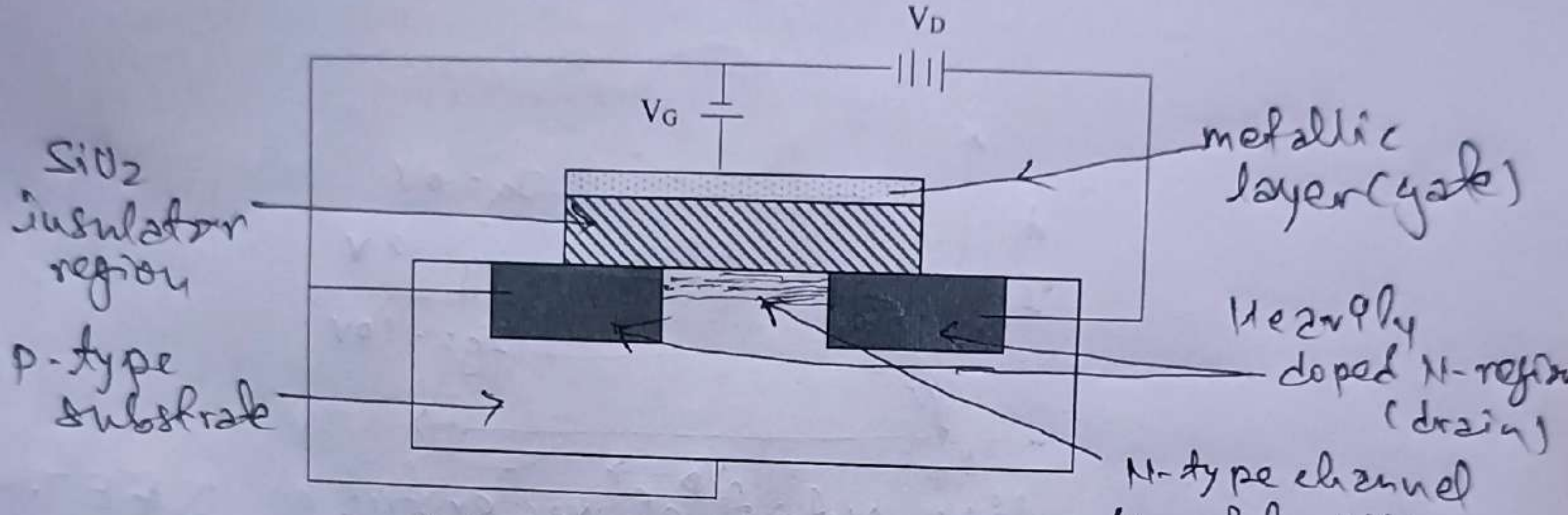


Fig: Structure and biasing of P-channel MOSFET.

When V_{GS} is zero no current can flow between source to drain. As a positive potential is applied at gate, this induced electron from p-substrate towards the sio_2 interface and hence a small n-type region will develop at interface due to applied field in gate. Then this conducting layer conduct between two heavily doped source and drain and current flows through the channel or substrate. More is V_{GS} potential larger will be width of channel thus, the large amount of current passing through MOSFET for a applied drain potential. Hence, in this way current can be controlled in MOSFT by applied gate potential.

The output characteristic curve of MOSFET is drawn between applied drain voltage and corresponding drain current through channel in MOSFET for a constant gate to source positive potential. On increasing drain potential corresponding drain current increases rapidly and becomes constant (saturate) at particular value known as pinch off voltage. Further, the value of saturation drain current also increases with increase in gate potential as shown in figure below.

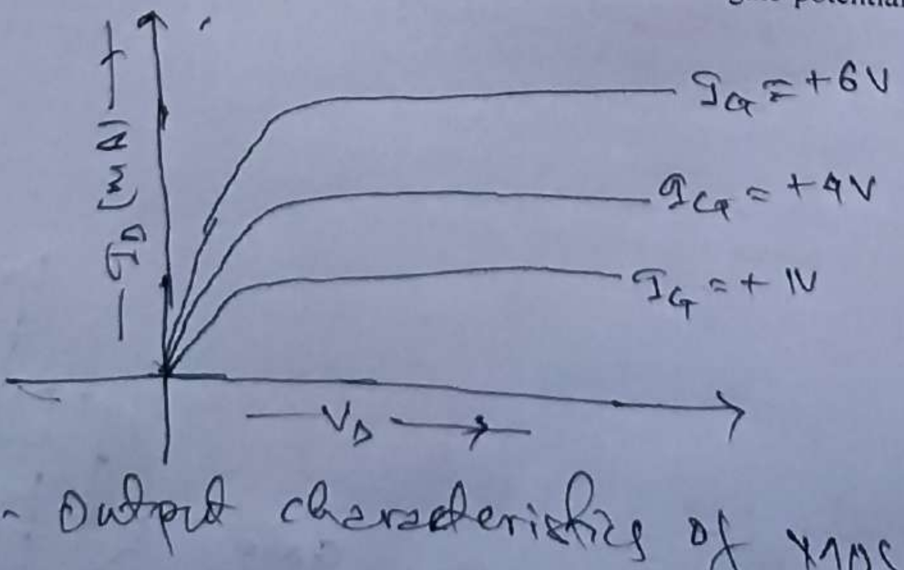


fig:- Output characteristics of MOSFET.